HDOC DAY-48

Title: Finding the Pivot Index in a Variable-Length Integer Array Using Pointers

1. Problem Statement

Write a C program that reads the size and elements of a variable-length integer array, arr[]. Pass the array to a function named pivot_index() using its address and pointers. The function pivot_index() must:

1. Calculate the left sum (sum of all elements strictly to the left of each index).
2. Calculate the right sum (sum of all elements strictly to the right of each index).
3. Treat the left sum $= 0$ for the first element.
4. Treat the right sum $= 0$ for the last element.
5. Find and display the leftmost index at which left sum $=$ right sum.
6. Display -1 if no pivot index exists.

2. Theoretical Background & Algorithm Logic

- **Pivot Index Search:**

First, the total sum of all elements in the array is computed. Then, the algorithm iterates across each index while maintaining a running left_sum.

- **Optimized Verification:**

The right_sum for any index is dynamically derived via total_sum - left_sum - arr[i]. If left_sum == right_sum matches, that index is immediately identified as the leftmost pivot index. If the loop completes without a match, -1 is returned.

3. Test Case Table

Test Case	Array Elements JPG+ 1	Expected Output JPG+ 1	Actual Output	Status
1	1 7 3 6 5 6	Pivot index: 3	Pivot index: 3	Pass
2	1 2 3	-1	-1	Pass
3	2 1 -1	Pivot index: 0	Pivot index: 0	Pass

4. Source Code Listing

```
#include <stdio.h>

void pivot_index(int *arr, int size) {
   int total_sum = 0;
```

```
    int left_sum = 0;

    for (int i = 0; i < size; i++) {
        total_sum += *(arr + i);
    }

    for (int i = 0; i < size; i++) {
        int right_sum = total_sum - left_sum - *(arr + i);

        if (left_sum == right_sum) {
            printf("Pivot index: %d\n", i);
            return;
        }

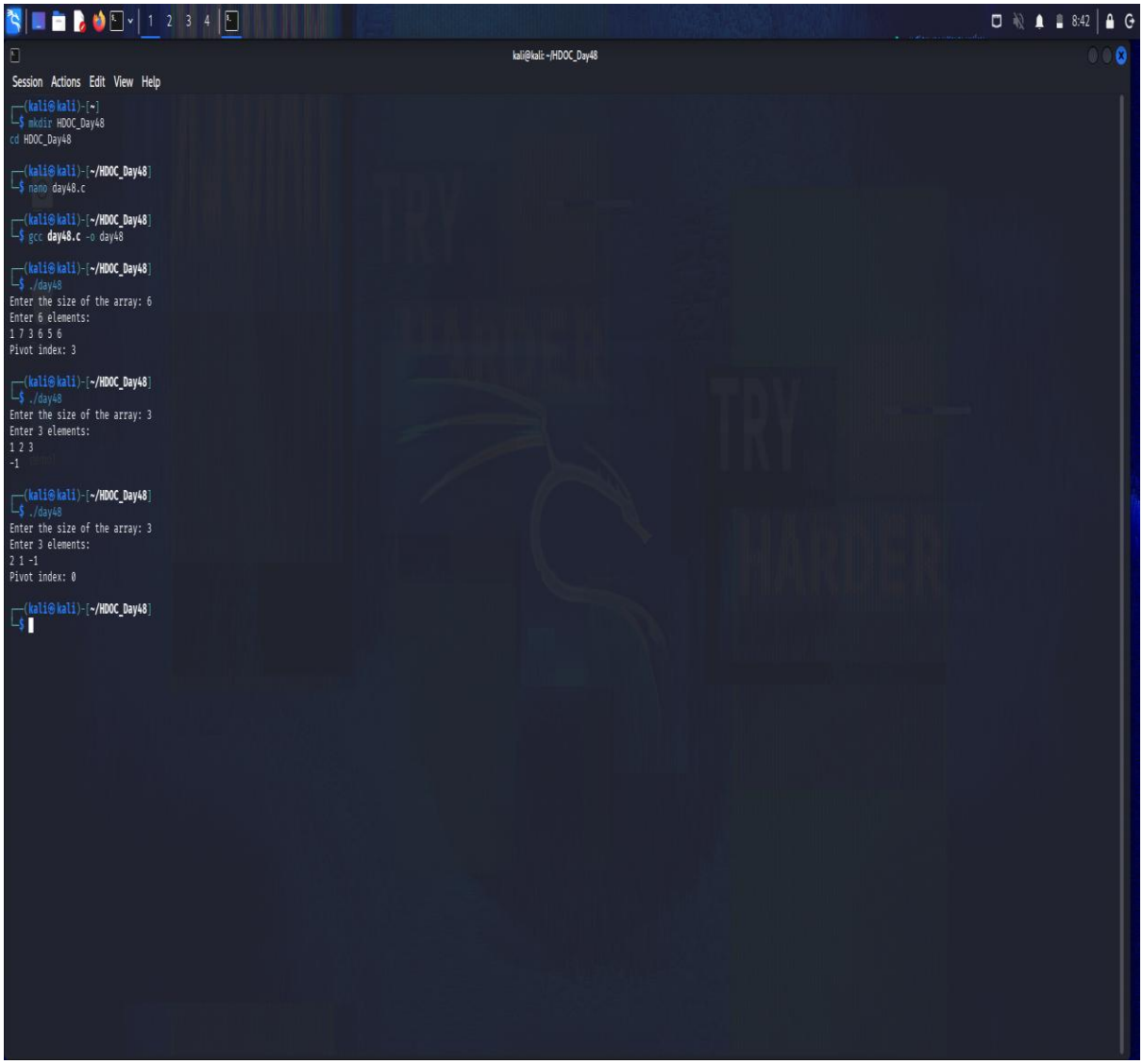
        left_sum += *(arr + i);
    }

    printf("-1\n");
}

int main(void) {
    int size;
    printf("Enter the size of the array: ");
    if (scanf("%d", &size) != 1 || size <= 0) return 1;

    int arr[size];
    printf("Enter %d elements:\n", size);
    for (int i = 0; i < size; i++) {
        scanf("%d", &arr[i]);
    }

    pivot_index(arr, size);
    return 0;
}
```


kali@kali: ~/HDOC_Day48
Session Actions Edit View Help
(kali㉿kali)-[~]
$ mkdir HDOC_Day48
cd HDOC_Day48
(kali㉿kali)-[~/HDOC_Day48]
$ nano day48.c
(kali㉿kali)-[~/HDOC_Day48]
$ gcc day48.c -o day48
(kali㉿kali)-[~/HDOC_Day48]
$ ./day48
Enter the size of the array: 6
Enter 6 elements:
1 7 3 6 5 6
Pivot index: 3
(kali㉿kali)-[~/HDOC_Day48]
$ ./day48
Enter the size of the array: 3
Enter 3 elements:
1 2 3
-1
(kali㉿kali)-[~/HDOC_Day48]
$ ./day48
Enter the size of the array: 3
Enter 3 elements:
2 1 -1
Pivot index: 0
(kali㉿kali)-[~/HDOC_Day48]
$